

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 08



By- Vishal Sir



Recap of Previous Lecture



Topic

Planar graphs



Topic

Region and degree of regions



Topic

Sum of degree theorem for regions



Topic

Simple planar graph



Topic

Connected planar graph and Euler's equation



Topic

Simple connected planar graph

Topics to be Covered



Topic

Polyhedral graph



Topic

Euler's equation for disconnected planar graph



Topic

Vertex coloring



Topic

Chromatic number





Topic : Euler's equation (For Connected Planar graph)

- For any connected planar graph G , $\left\{ \begin{array}{l} \text{may or may not be} \\ \text{a simple graph} \end{array} \right\}$ following Equation will hold true

$$|V| + |R| = |E| + 2$$

← and it is called Euler's Equation for connected planar graph

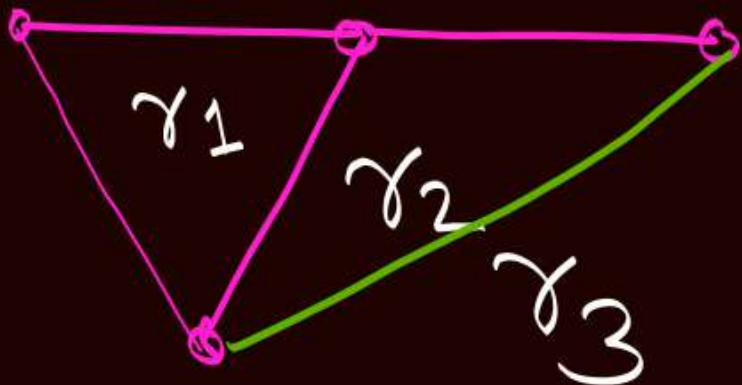
Where,
 $|V|$ = No. of vertices
 $|R|$ = No. of regions
 $|E|$ = No. of edges

$|V| + |R| = |E| + 2$ Will not hold true if Planar graph is disconnected

Some author define the Euler's equation using the following format

$$n + f - e = 2$$

Where
 n = No. of vertices
 f = No. of faces
 e = No. of edges



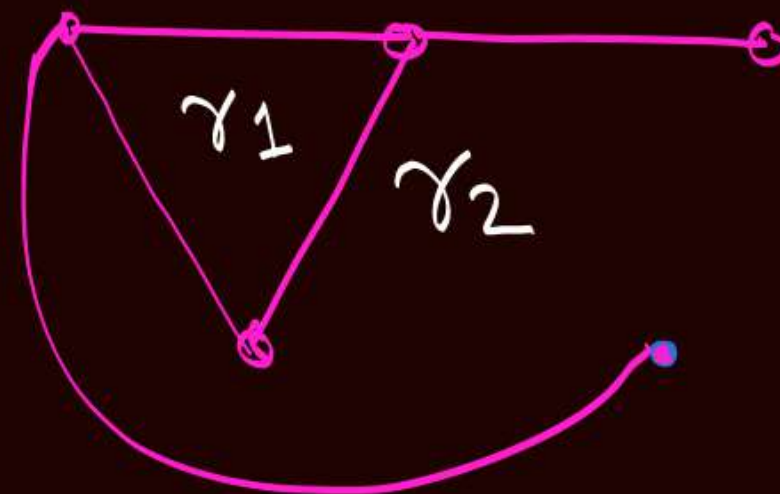
$$|V| + |R| = |E| + 2$$

$$4 + 2 = 4 + 2$$

$$+0 \quad ? = +1 \quad +1$$

✓

✓



$$|V| + |R| = |E| + 2$$

$$4 + 2 = 4 + 2$$

$$+1 \quad ? = 0 \quad +1$$



Topic : Polyhedral graph

A polyhedral graph is a simple connected planar graph in which degree of each 'vertex' is at least '3'.

∴ $3|V| \leq 2|E|$ will also true along with all '4' equations of simple connected planar graph

Let

$$\boxed{3|V| \leq 2|E|} \text{ is eqn-⑤}$$

• In a polyhedral graph, following equations will hold true

$$3|R| \leq 2|E| \text{ ——— ①}$$

$$|V| + |R| = |E| + 2 \text{ ——— ②}$$

$$|E| \leq 3|V| - 6 \text{ ——— ③}$$

$$|R| \leq 2|V| - 4 \text{ ——— ④}$$

$$3|V| \leq 2|E| \text{ ——— ⑤}$$

From eqⁿ ⑤ $3|V| \leq 2|E| \therefore |V| \leq \frac{2|E|}{3}$

Put this value of $|V|$ in Euler's eqⁿ

ie, $\frac{2|E|}{3} + |R| \geq |E| + 2$

$$|R| \geq \frac{|E|}{3} + 2 \Rightarrow |E| \leq 3|R| - 6$$

————— eqⁿ ⑥

From eqⁿ ⑤ $3|V| \leq 2|E| \therefore |E| \geq \frac{3|V|}{2}$

Put this value of $|E|$ in Euler's equation

ie, $|V| + |R| \geq \frac{3|V|}{2} + 2$

$$|R| \geq \frac{|V|}{2} + 2$$

$$|V| \leq 2|R| - 4 \text{ ——— eqⁿ ⑦}$$

Note:-

In a polyhedral graph all seven equations from eqⁿ ① to eqⁿ ⑦ will hold true

Euler's equation for disconnected Planar graph

may or may not?
be simple

Let 'G' is a planar graph With 'K' Connected Components.

Let $|V|$ = Total no. of vertices in G.

$|R|$ = Total no. of regions in G

$|E|$ = Total no. of edges in G

K : No. of Connected Components in G.

Graph G with 'K' Connected Components

1st
Connected
Component

2nd
Connected
Component

3rd
Connected
Component

(K-1)st
Connected
Component

Kth
Connected
Component

let $n = |V_1|$

$e = |E_1|$

$f = |R_1|$

$|V_1| + |R_1| = |E_1| + 2$

$|V_2|$

$|E_2|$

$|R_2|$

$|V_2| + |R_2| = |E_2| + 2$

$|V_3|$

$|E_3|$

$|R_3|$

$|V_3| + |R_3| = |E_3| + 2$

$|V_K|$

$|E_K|$

$|R_K|$

$|V_K| + |R_K| = |E_K| + 1$

$$|V_1| + |R_1| = |E_1| + 2$$

$$|V_2| + |R_2| = |E_2| + 2$$

$$|V_3| + |R_3| = |E_3| + 2$$

⋮

⋮

$$|V_k| + |R_k| = |E_k| + 2$$

$$\underbrace{(|V_1| + |V_2| + \dots + |V_k|)}_{|V|} + \underbrace{(|R_1| + |R_2| + \dots + |R_k|)}_{|R| = ? \text{ Wrong}} = \underbrace{(|E_1| + |E_2| + \dots + |E_k|)}_{|E|} + 2k$$

$$\underbrace{|R| + (k-1)}_{\text{Correct}}$$

$$\therefore |V| + |R| = |E| + 2k - (k-1)$$

$$\Rightarrow |V| + |R| = |E| + k + 1$$

Euler's equation for disconnected Planar graph

{ may or may not?
be simple }

Let 'G' is a planar graph with 'K' Connected Components.
{ may not be simple }

then following Equation will hold true

$$|V| + |R| = |E| + K + 1$$

it is called
Euler's Equation
for disconnected planar graph

Where,

$|V|$ = No. of Vertices in G

$|E|$ = No. of Edges in G

$|R|$ = No. of regions in G = (Total no. of interior regions + '1' Exterior region)

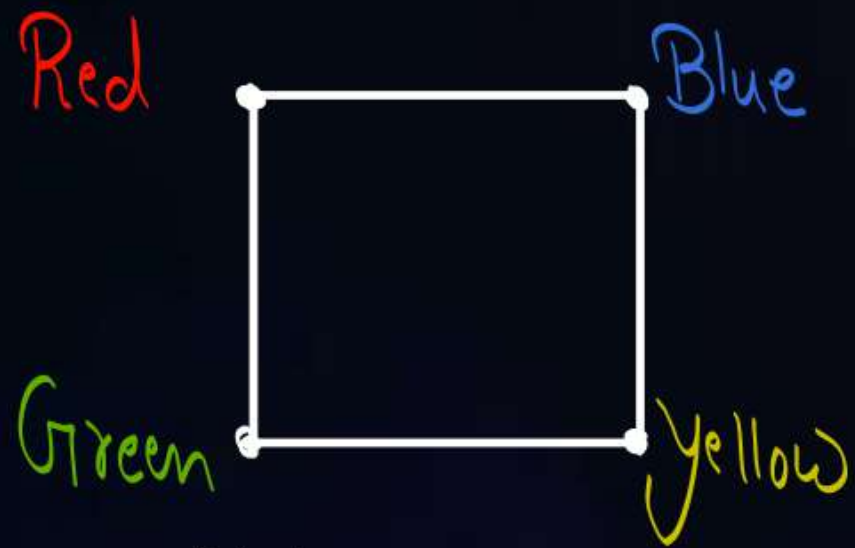
K = No. of Connected Components in G

Coloring

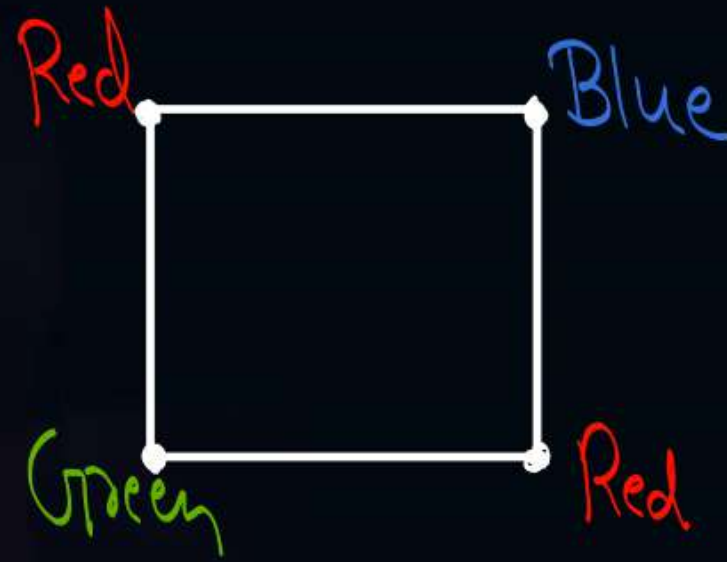


Topic : Vertex coloring

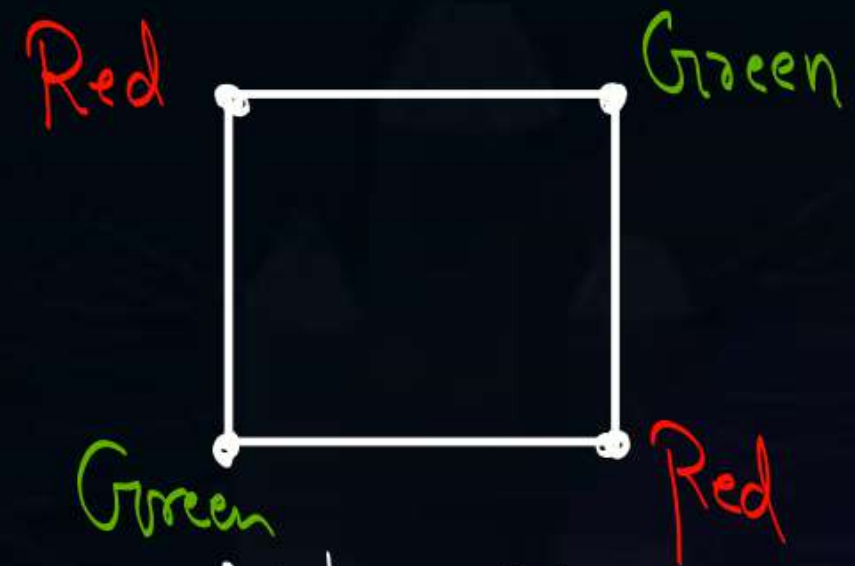
- An assignment of colors to the vertices of graph G , such that no two adjacent vertices of the graph have the same color is called vertex coloring of graph G .



Vertex Coloring
Using '4' Colors



Vertex Coloring
Using '3' Colors



Vertex Coloring
Using '2' Colors

All of them are Valid Vertex Coloring of given graph

→ If we have a graph with ' n ' vertices, then we can always vertex color that graph using ' n ' colors { i.e. one color per vertex }

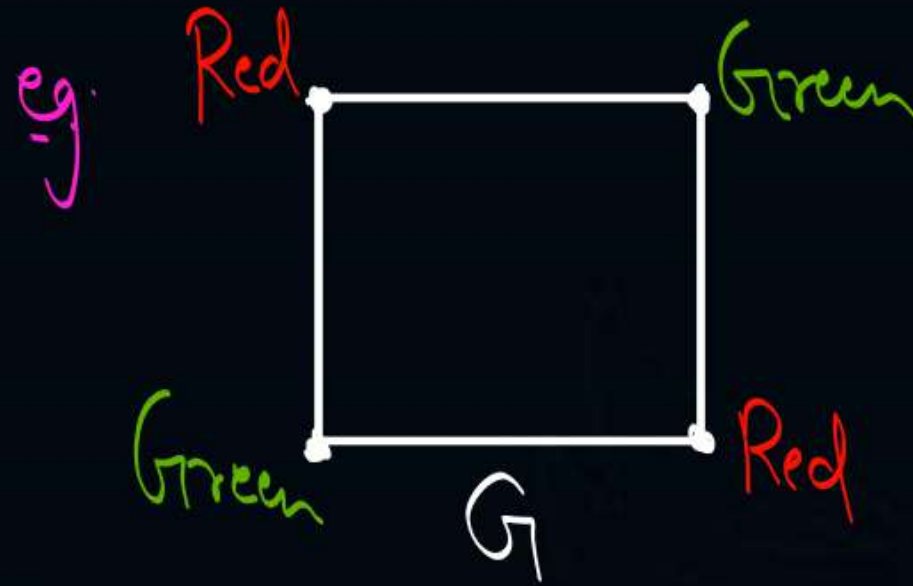
"But we are interested in the minimum number of colors required for the vertex coloring of the given graph"



Topic : Chromatic number

- Minimum number of colors needed for the vertex coloring of graph G is called chromatic number of graph G,

It is denoted by $\chi(G)$.



We can not vertex color the graph G using less than '2' colors
∴ $\chi(G) = 2$



Topic : K-colorable vs K-chromatic

K-Colorable

* If there exist a vertex coloring of graph 'G' that uses 'K' Colors, then graph G is called 'K' Colorable

if graph G is K-Colorable then,
 $\chi(G) \leq K$

K-Chromatic

If Chromatic number of graph 'G' is 'K', then graph 'G' is said to be K-chromatic

* i.e. Graph G is K-chromatic if and only if
 $\chi(G) = K$

Note:- *

① If graph G is K -Colorable, then graph G is $(K+1)$ -Colorable as well,

Provided $(K+1) \leq$ No. of vertices in graph G .

② If graph G is K -Colorable, then graph G may or may not be $(K-1)$ -Colorable

③ If graph G is K -chromatic, then it is neither $(K+1)$ -chromatic nor $(K-1)$ -chromatic



Topic : Four-Color Theorem



→ Every planar graph G is four colorable.

• For any planar graph G , $\chi(G) \leq 4$

i.e. there exist at least
vertex coloring that
uses '4' colors

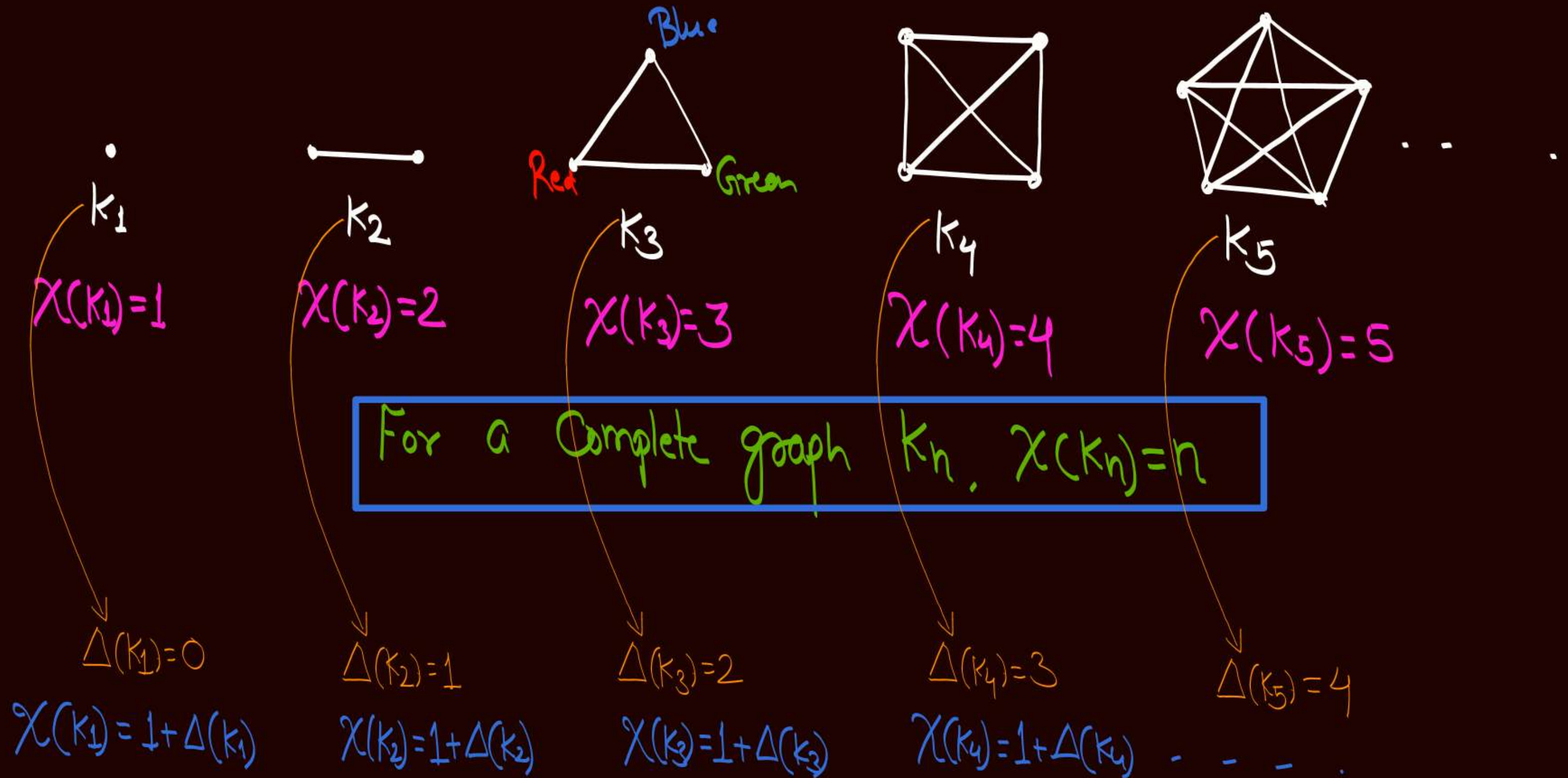
Note:-→

① Chromatic number of graph G
can be '1' if and only if
graph ' G ' is a NULL graph

i.e. $(\chi(G)=1) \iff 'G' \text{ is NULL graph}$

② If graph G is not a NULL graph
then $\chi(G) \geq 2$

Q:- Find the Chromatic number of Complete graph ' K_n '

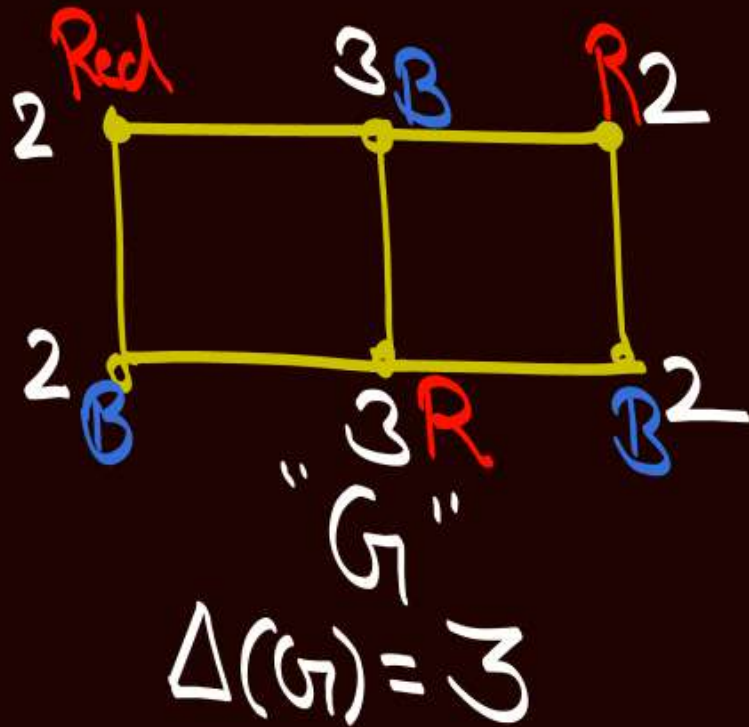


For any graph G ,

$$\chi(G) \leq 1 + \Delta(G)$$

{ in previous example
 $\chi(G) = 1 + \Delta(G)$ }

eg.



And we can observe
 $\chi(G) = 2$

{ in this example
 $2 < 3$
i.e. $\chi(G) < \Delta(G)$ }

$$\chi(G) \geq \delta(G)$$

Wrong Concept.
"need not be true"

HW Q:- Find the Chromatic number of Cycle graph ' C_n '

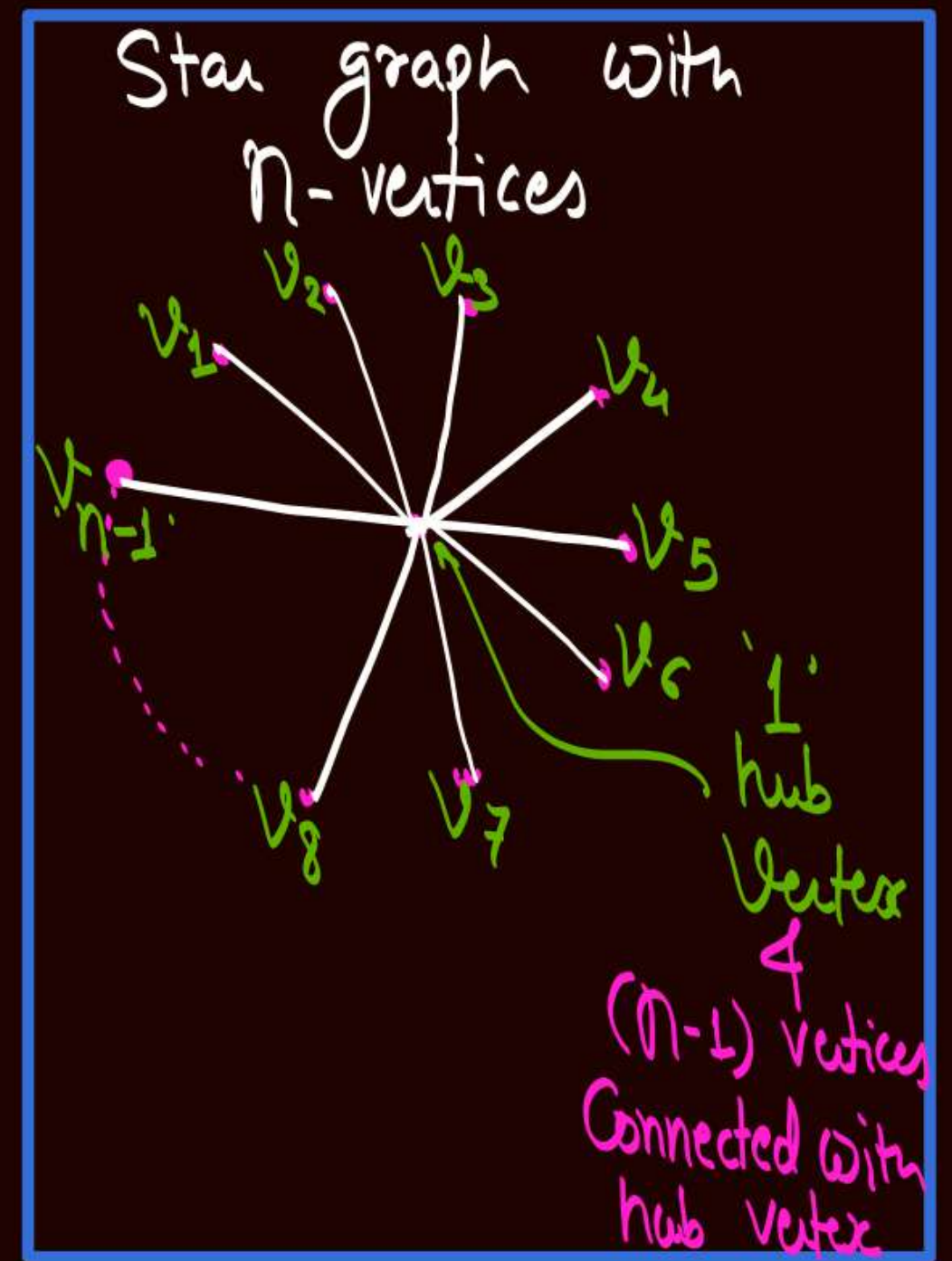
$$\chi(C_n) = \begin{cases} 2, & \text{when } n = \text{even} \\ 3, & \text{when } n = \text{odd} \end{cases}$$

H.W. Q:- Find the Chromatic number of Wheel graph 'W_n'

$$\chi(W_n) = \begin{cases} ? & \text{when } n = \text{Even} \\ ? & \text{when } n = \text{odd} \end{cases}$$

H.W.
Q:- Find the Chromatic number of Complete bipartite graph $K_{m,n}$

H.W
Q:- Find the Chromatic number of a star graph with 'n' vertices





2 mins Summary



Topic

Polyhedral graph

Topic

Euler's equation for disconnected planar graph

Topic

Vertex coloring

Topic

Chromatic number

THANK - YOU